Climatic variables and the reemerging Eastern and Venezuelan Equine encephalitis in equines, Colombia 2005-2023

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Summary

Background. Eastern Equine encephalitis (EEEV) and Venezuelan equine encephalitis (VEEV) are vector-borne diseases of public health concern that are likely to reemerge or spread due to climate change**. Methods.** Between 2005 and July 2023, a retrospective cross-sectional descriptive observational study was conducted to analyze the cases of EEEV and VEEV in Colombia. Climatic variables of temperature, rainfall, and El Niño and La Niña phenomena were analyzed. The reports of encephalitis cases were obtained from the Minister of Agriculture. Descriptive statistics were performed, and a Pearson correlation coefficient was used**. Findings.** 169 cases of EEEV and 102 of VEEV in horses were found; the cases remained low between 2005-2015. In 2016, there was a strong epizootic of EEEV (n=57 cases) in the department of Casanare bordering Venezuela. That same year, there was a significant increase in VEEV cases (n=15 cases) in the department of Cesar on the border with Venezuela. Between 2017 and 2019, there was a decrease in both cases of encephalitis, but in 2020, another essential epizootic appeared, with a peak of 31 cases of EEEV in the department of Casanare. The climatic variables and the phenomena of La Niña and El Niño are not associated with the cases of VEEV and EEEV. **Interpretation**. The solid epizootic activity of the EEEV and VEEV viruses shows that both viruses can become human and animal public health problems. Based on the present work, neighboring countries must be alerted to the increased encephalitis cases. **Funding.** None.

Keywords: Reemerging infectious diseases, arthropod borne viral encephalitis, agricultural workers diseases, environment, preventive medicine public health, epidemiology, horse diseases, Tropical Climate, El Nino-Southern Oscillation.

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Introduction

Viruses transmitted by arthropods cause important encephalitis; several equine encephalitis viruses belong to the family *Togaviridae* genus *Alphavirus*. In addition to alphavirus Chikungunya and Mayaro, there are Western equine encephalitis viruses (WEEV), Venezuelan equine encephalitis virus (VEEV), and Eastern equine encephalitis virus (EEEV). [1] EEEV was previously classified into four distinct lineages/subtypes (I–IV). South American strains (lineages II-IV) include isolates from South and Central America; strains are genetically distinct from those from North America that belong to lineage I. Due to genetic divergence and significant differences in ecology and pathogenesis, South American isolates of EEEV were recently classified as a distinct species, as the Madariaga virus (MADV). [1]

EEEV and VEEV cause critical febrile illness and encephalitis in the Americas. The VEEV virus is part of the virus species in the VEEV complex. [2] The enzootic viral species within the alphavirus genus are the Mosso das Pedras, Everglades, Mucambo, Tonate, Pixuna, Cabassou virus, and the Rio Negro virus. [2] Most cause feverish symptoms of encephalitis in humans and horses. [3] Most encephalitis viruses have wild rodents as the main enzootic vertebrate reservoirs; from this enzootic step, the mosquitoes transmit the encephalitis viruses to the equids. Finally, it becomes an epizootic from the equids, ending in the accidental human host. It is also important to remember the role of various mammals in the transmission chain of encephalitis, which has been documented, especially VEEV and EEEV. [4]

In the Latin American tropics, the abundant species Culex (*Melanoconion*), *Sorophora*, *Aedes*, *Coquillettidia*, *Culiseta melanura* are the main mosquito vectors of encephalitis viruses and human disease. [5] Encephalitis viruses, especially Togavirus and Flavivirus, have in common an unexpected abrupt increase in incidence in equine populations. EEEV and VEEV encephalitis are considered the most severe emerging viral diseases. Encephalitis has a wide distribution throughout the Americas and the Caribbean. [3,6]

In 1938 the United States, human mortality rates of 35% were reported. In horses, it usually has mortality rates between 70% to 90%. Infection in humans and horses requires invertebrate vectors that feed on birds and mammals, such as *Aedes*, *Coquillettidia*, or Culex. [7]

Equines act as sentinels or indicators of the beginning of an epidemic; in a natural focus, they tend to be the first to manifest disease. [5] In Colombia, the EEEV and EEEV viruses are notifiable; encephalitis is distributed throughout the country, especially in the Caribbean and jungle regions.[8] In Colombia, equine surveillance is carried out by the Colombian Agricultural Institute (ICA), and a significant increase has been noted in the last decade, [8-10] especially in the departments dedicated to livestock production. In human populations, Gil et al.[11] found a seroprevalence of 4.4% of VEEV in the department of Cauca in southern Colombia in the Andean area. This demonstrates the sizeable viral circulation favored by the abundance of *Culex* mosquito vectors widely distributed throughout the territories.

On the other hand, climatic variables have been associated with increased emerging and reemerging diseases and vector-borne diseases, especially arboviral diseases. Studies have shown an increase in malaria, dengue, and encephalitis. [12] Global climate variations may play a role in

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increasing outbreaks of vector-borne infectious diseases due to heavy rainfall and intense warming trends that can turn additional regions into suitable habitats for pathogen vectors. [13] The increase in extreme climatic conditions can generate problems that, with the increase in some arboviral diseases, create the need to evaluate the climatic anomalies that can cause the development of future outbreaks. [14] Studies in Florida show that EEEV activity in horses was associated with more significant amounts of rainfall during the month of most extraordinary transmission, as well as the previous three months. Horse cases were lower during El Niño winters but higher during the following summer. In contrast, La Niña winters were associated with more cases and fewer during the following summer.[15] Deficient precipitation levels were associated with suppressing EEEV cases for three months. Given the periodicity and potential predictability of El Niño Southern Oscillation (ENSO) cycles, precipitation, and temperature may provide a method to predict the potential risk of EEEV. In Florida, the distributed lag nonlinear model's results effectively predicted general patterns of EEEV transmission. ENSO predictors of EEEV could be used with previously validated EEEV risk mapping, in conjunction with current serological surveillance of mosquito and chick abundance, to assist in targeted prevention strategies that reduce the potential for EEEV transmission. [15] The present work is the first in Colombia to get information about encephalitis and its relationship with climate variables.

The objective of this study was to carry out a retrospective analysis of the cases of EEEV and VEEV in equines and their relationship with climate variables between 2005 and June 2023.

Methods.

Type of study and geographic location. Between 2005 and July 2023, a retrospective cross-sectional descriptive observational study was conducted in Colombia. Colombia is in the northeast corner of South America. The country is crossed from South to North by the Andes mountains; the lowland plains bordering Venezuela are on the East. To the West are the geographical areas parallel to the Pacific Ocean, bordering Panama, and to the south, the Amazon region, bordering Brazil, Peru, and Ecuador. To the country's North is the Caribbean area with a marked tropical climate with rainy and summer seasons. [16] The departments that presented one or more cases of EEEV and VEEV were selected; 24/32 (75%) of the departments of the country that presented cases of EEEV and VEEV were included, all the cases presented in the five large geographical regions of Colombia.

Obtaining data on encephalitis. The diagnosis of EEEV and VEEV encephalitis in horses was made by the Colombian Agricultural Institute (ICA), which is attached to the Ministry of Agriculture of Colombia. The ICA is the only one authorized to diagnose encephalitis, rabies, leptospirosis, avian flu, brucellosis, and tuberculosis. The report of cases by the department between 2005 and July 2023 was obtained through the ICA epidemiological bulletins.[8,9]

Analysis of climatic variables. Three climatic variables were analyzed: temperature, rainfall, and the Niño and Niña meteorological phenomena. Between 2005 and July 2023, the Oceanic Nino Index (ONI) meteorological data were obtained from the National Oceanic and Atmospheric Administration Climate (NOAA).[17] The ONI is calculated as the three-month moving average of sea surface temperature anomalies for the El Niño 3.4 region (5°N-5°S, 120° -170°W). Positive

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anomalies occur when the ONI value reports a threshold greater than 0.5 (El Niño). Negative anomalies refer to values less than -0.5 (La Niña). Values between the range of -0.5 and 0.5 are normal event conditions.[17] Temperature and rainfall data from 2005 to 2022 were obtained from the Colombian Institute of Hydrology, Meteorology, and Environmental Studies (IDEAM). (18) Of the 24 departments that reported EEEV or VEEV, the averages of the maximum and minimum temperatures and the annual rainfall reported by the IDEAM weather stations were obtained. In Colombia, geographic and climatological differences exist in the regions mentioned above (Andes, Amazonia, Llanos, Pacific, and Caribbean).[16]

Statistical analysis. Climate data for precipitation and temperature were pooled annually; EEEV and VEEV cases were grouped by month, year, and department. A correlation was made between the ONI scale and the cases of both encephalitis. In addition, the association was made between the variable's precipitation or temperature concerning EEEV and VEEV. Descriptive statistics were performed using the statistical package v4.3.0 in R v.3.5.0 59. About the correlation, hypotheses for the significance of the correlation coefficient were tested. In which the p-value was set to 0.05. To quantify the linear relationship between two variables. That is, the analysis makes it possible to establish whether the change in one variable is associated with the change in the value of the other. For this reason, a scale between -1 and 1 is used for interpretation.[19]

Results

Between 2005 and July 2023, 169 cases of EEEV and 102 of VEEV were found. In 2016, 39% (n=66) of the total cases of EEEV and 21.6% (n=22) of VEEV were reported. Cases remained at low levels between 2005-2015 (Figure 1). However, in 2016, there was a significant epizootic of EEEV (n=57 cases) in the department of Casanare bordering Venezuela. That same year, there was a significant increase in VEEV cases (n=15 cases) in the department of Cesar on the border with Venezuela (Figures 1, 2). During the study period, the departments with the most reports of VEEV were Casanare (n=95), Meta (n=15), and Arauca (n=11). VEEV was mainly recorded in Cesar (n=21), Cordoba (25), and Magdalena (n=15).

The association between EEEV and VEEV concerning the ONI is shown in Figure 3. The El Niño and La Niña phenomena are identified when the ONI indices surpass 0.5 or fall below -0.5, respectively. Therefore, in 2011, a strong Niña phenomenon was reported, a period in which there were no significant cases of both encephalitis. In contrast, between 2015-2016, there was a very intense El Niño phenomenon and an increase in cases of both encephalitis in the second semester of 2016 (Figure 3).

The analysis of the correlation of the monthly records for both variables (EEEV vs. ONI) was conducted, employing the Pearson correlation coefficient (r=-0.101, p=0.135). The derived p-value exceeding 0.05 leads us to conclude that no statistically significant correlation exists between the EEEV and ONI variables. A similar analysis between VEEV and ONI revealed a lack of correlation (r=0.033, p=0.622). Additionally, the statistical association by year between the climatic variables (temperature and precipitation) and the cases of EEEV and VEEV demonstrated no correlation (Table 1).

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Table 1. Correlation coefficients of the association between temperature, rainfall, and the cases of EEEV and VEEV in Colombia.

Variables		Pearson's correlation	P value
Cases of EEEV	Temperature	0.41	0.09
	Rainfall	0.25	0.31
Cases of VEEV	Temperature	0.23	0.36
	Rainfall	0.23	0.35

*Correlation is significant at the 0.05 level

Discussion

The results of the present work demonstrate the danger to public health that Eastern and Venezuelan encephalitis represents in Colombia. The epidemiological behavior of EEEV in recent years is remarkable; its recent activity is the beginning of a reemergence that could affect more horses and jump to humans. A retrospective study by Bonilla et al. [10] between 2008 and 2019 found a higher risk of epizootics for Venezuelan encephalitis in the departments bordering Venezuela. According to our analysis, the trend is that the activity is more significant with EEEV, and apparently, there is a decrease in VEEV. This pattern suggests that VEEV is endemic in Colombia and appears less influenced by outbreaks in border areas, unlike EEEV. Remember that there is no approval in Colombia vaccine against EEEV. The effectiveness of the vaccination campaigns carried out by the ICA could explain the extensive periods of low frequency of VEEV outbreaks in recent years. The low infection rates by both viruses in the bordering departments with Venezuela, such as La Guajira, Vichada, and Guainia, are striking, especially in La Guajira, close to Venezuela, where we expected more cases of VEEV.

Also, in departments bordering Brazil, Peru, and Ecuador, such as Vaupes, Amazonas, and Putumayo, we expected more cases of EEEV (Figure 2). Unfortunately, both encephalitis are not routinary studied in human patients. Under the shadow of other arboviruses such as Dengue, Yellow Fever, Zika, Mayaro, and Chikungunya, equine encephalitis could be masked and play an essential role in cases of human encephalitis in departments with more EEEV and VEEV cases. It is, therefore, necessary, based on the results of this study, to inform the medical community about the possible etiology that both viruses could have in cases of human encephalitis. [1,6] Additionally, the emergence of focused outbreaks of EEEV in departments in the interior of the country, such as Quindío, highlights the urgent need to investigate, under the One Health vision, the contributing and conditioning factors in these specific areas, contemplating the existence of environmental, ecological and sociodemographic factors. That could be directing the risk and distribution of these diseases. The occurrence and frequency of EEEV and VEEV in different departments have profound epidemiological implications, especially considering interdepartmental mobility and proximity to other countries. The variability observed in the frequencies points to the influence of various factors that could significantly impact the distribution of these diseases.

The growing worldwide importance of EEEV and VEEV as zoonoses in Neotropical America has been demonstrated by publications from 1925 to 2019. [12, 20] The distribution of EEEV and VEEV in countries bordering Colombia and the rest of South America is depicted in Figure 4. Clinical EEEV disease in non-human hosts has been demonstrated in horses, birds, camelids, deer, dogs,

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and pigs. VEE's main reservoirs are rodent species *Sigmodontinae*; [6] bats can also be accidental reservoirs of VEEV. [31] Birds are also susceptible to EEEV; there are no surveillance studies in Colombia on chickens, so there is an economic risk for poultry farmers. [12] Horses are susceptible to EEEV, and 70-90% die, with those that survive having permanent brain damage. In humans, EEEV has a high fatality rate (33%), and surviving patients develop severe sequelae; no vaccine or antiviral treatment is approved for humans. [12]

On the other hand, it is essential to keep in mind that the clinical symptoms of encephalitis are very similar to dengue, Zika, and Chikungunya, which is why it is likely that many cases of encephalitis are not adequately diagnosed. Furthermore, there are no routine laboratory tests for diagnosing encephalitis, as there are for Dengue, Zika, and Chikungunya. [32]

Seroepidemiological studies of the 2010 outbreak in Panama with serum samples from individuals from the same house as the confirmed cases, as well as from the cases of horses that presented neurological signs, were analyzed using the hemagglutination inhibition and neutralization test by reduction of plate (Figure 4). The results indicated that the seroprevalence for MADV in humans was 19.4% (Figure 4), VEEV 33.3%, and Mayaro virus 1.4%. Compared with individuals aged 2 to 20, older individuals (21≥41 years) were five times more likely to have VEEV antibodies, while the prevalence rate of MADV was independent of age. The overall seroprevalence of MADV in equids was 26.3%, VEEV 29.4%, West Nile virus (WNV) 2.6%, and St. Louis encephalitis virus (SLEV) 63%.[23] The work demonstrated that multiple arboviruses are circulating in human and equine populations in Panama and South America as well (Figure 4).

It is very likely that in Colombia, there have been cases of encephalitis by EEEV and VEEV that have been mistakenly classified as dengue, Zika, or Chikungunya cases. Now, there is an intense outbreak of Dengue in Colombia and South America, and no one could be thinking about encephalitis. Misdiagnosis of VEEV as dengue is primarily due to the overlapping complement of signs and symptoms associated with typical infections. [1,2] However, endemic VEEV cases are much more likely to be misdiagnosed than epidemic cases that occur during ongoing equine outbreaks because the latter alert public health officials to the possibility of VEEV involvement. Therefore, endemic VEEV has received little attention, and control measures targeting the Culex (*Melanoconion*) vector species transmitted from rodents to humans are rarely instituted. [1, 33]

Despite confirmatory diagnostic tests for EEEV and VEEV, including virus isolation, PCR, and serological analyses, many cases are only diagnosed clinically and are therefore confused with other febrile illnesses. The lack of a confirmatory diagnosis is primarily due to the costs associated with diagnosis in resource-poor regions. [2, 34] The detection of human cases of VEEV in Bolivia, which occurred after the implementation of febrile surveillance activities, demonstrated the need to establish this kind of work in other South American countries. [35]

Estimating the impact of VEEV and EEEV infection on public health in Latin American countries remains a challenge. [32] Although encephalitis is endemic in many areas of Colombia, reports of neurological sequelae and mortality associated with encephalitis are sporadic. This is likely due to the limitations of current surveillance systems that do not perform follow-up studies. [1,6,36]

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Encephalitis is endemic in the Latin American tropics. In South America (Colombia, Venezuela, Brazil, Ecuador, Peru, Uruguay, and Argentina), there are studies on VEEV and EEEV in mosquitoes, horses, humans, and several mammals. Studies show that Eastern and Venezuelan encephalitis is a latent public health problem that has not received the adequate attention it deserves as an emerging viral disease. [2,6] Epidemics caused by epizootic strains of VEEV were first recognized during the 1920s and have occurred periodically in northern South America, Central America, and Mexico, involving hundreds of thousands of people and equids; later, EEE, WNV, and SLEV were reported. [1,6]

On the other hand, regarding the climatological findings, the analysis showed no statistical associations between the cases of EEEV and VEEV (Table 1). In contrast, Miley et al. [15] found statistical associations of EEE with precipitation, cooling degree days, and Southern Oscillation Index. According to their analysis, these findings predict EEE activity in horses. Their analysis was more robust than ours, as they performed more complex nonlinear regression mathematical models applied to transmission dynamics in Florida, United States. In addition, the study in Florida was carried out with 730 cases; [15] in contrast, with 271 cases in our work, the more significant difference in EEEV cases could influence the absence of statistical associations in the present study.

Although Figure 3 shows an increase in EEEV cases during the El Nino phenomenon, it is a visual effect since, statistically, it does not seem to be so (Table 1). If the temperatures, the rains, and the Nino and Nina phenomena did not influence the equine encephalitis in this study, we could infer that the epizootics would be influenced by the vectors and not by the weather. However, changes in the availability of resources, vegetation cover, and water bodies also influence variables not analyzed in the present work. [37] However, if the vectors depend on the climate, it is possible to hypothesize that mosquitoes adapt to survive in the harsh Colombian tropics. The climatic influence on vector dynamics is evident; on the island of Cyprus in the Mediterranean, a study was conducted where the most numerous mosquito species (*Cx. pipiens, Ae. detritus,* and *Ae. caspius*) were correlated concerning climatic variables. [38] The results showed that captured mosquitoes were not statistically associated with temperature, similar to what we found, but with encephalitis cases.

The moderate increase in VEEV could be due to vaccination; however, the low casuistry in the departments of La Guajira, Cesar, Arauca, Vichada, Casanare, Vaupes, and Guania, bordering Venezuela and Brazil, is striking. Due to the difficulties of access to the departments of Arauca, Vichada, Casanare, Vaupes, and Guania, the surveillance of encephalitis cases is not systematic like the rest of Colombia. Based on the results of this study, both encephalitis case studies and VEEV vaccination could be focused and intensified (Figure 2).

On the other hand, the public health crisis in Venezuela due to sociopolitical problems has allowed an increase in cases of vector-borne diseases. [39] In this sense, Colombia, its close neighbor, must intensify entomological surveillance and encephalitis cases to control possible outbreaks that may arise in time.

According to the FAO, Colombia is the ninth country with the largest horse population in the world.[40] The equine population has increased in recent years. According to the 2013 national census, there was a population of 1,553,157 horses, and in 2022, 1,600,415.[41,42] Equines are

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distributed mainly in the departments of Antioquia (15.9%), Córdoba (6.8%), Tolima (6.3%), Cauca (6.1%), Cundinamarca (6.1%), Bolívar (5.6%), Casanare (5.4%), and Meta (5.4%).[41] Notably, the main departments with the equine population found the highest cases of EEEV and VEEE in this work. However, Casanare, which has less equine population, was the department that showed the most significant activity of both encephalitis. In this context, the importance of carrying out entomological studies is highlighted, which are scarce in Colombia for arboviruses other than dengue. The above will allow us to evaluate the health situation concerning encephalitis in these regions, considering the importance of horses as amplifiers and transmitters of the virus.

In conclusion, the solid epizootic activity of the EEEV and VEEV between 2005-2023 shows that both viruses can become a human and animal public health problem. EEEV and VEEV are vector-borne diseases of public health concern that are likely to reemerge or spread due to climate change. Technical preparation is essential to comprehensively address the anticipated risk of increased EEE cases. Misdiagnosis of EEEV and VEEV, such as dengue, Zika, or Chikungunya, is mainly due to the overlapping complement of signs and symptoms associated with characteristic infections. The remarkable activity of EEEV and VEEVE requires vigilance in the vector mosquitoes *Culex* sp and *Coquililletidia*, among others, to anticipate the risk of a possible epizootic by *Togaviridae* and *Flaviridae*, as happened with Chikungunya and Zika.[1] Based on the present work, countries such as Panama, Venezuela, Brazil, Peru, and Ecuador also must be alert to the increase in encephalitis cases in Colombia. The possible influence of proximity to border areas implies an urgent need to intensify cross-border cooperation and epidemiological surveillance to mitigate the transmission of these and other priority pathogens between nations, thus minimizing the risks of epidemics. Understanding epidemiological patterns and contributing factors is crucial to formulating more accurate and practical strategies to manage and control these encephalitis, thus safeguarding animal and public health in the affected regions. Under the One Health concept, the National Institute of Health could articulate with the Colombian Agricultural Institute and begin human and mosquito surveillance to prevent a possible outbreak.

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Figure 1. Cases of EEEV and VEEV, annual rainfall, and temperatures, 2005-2022

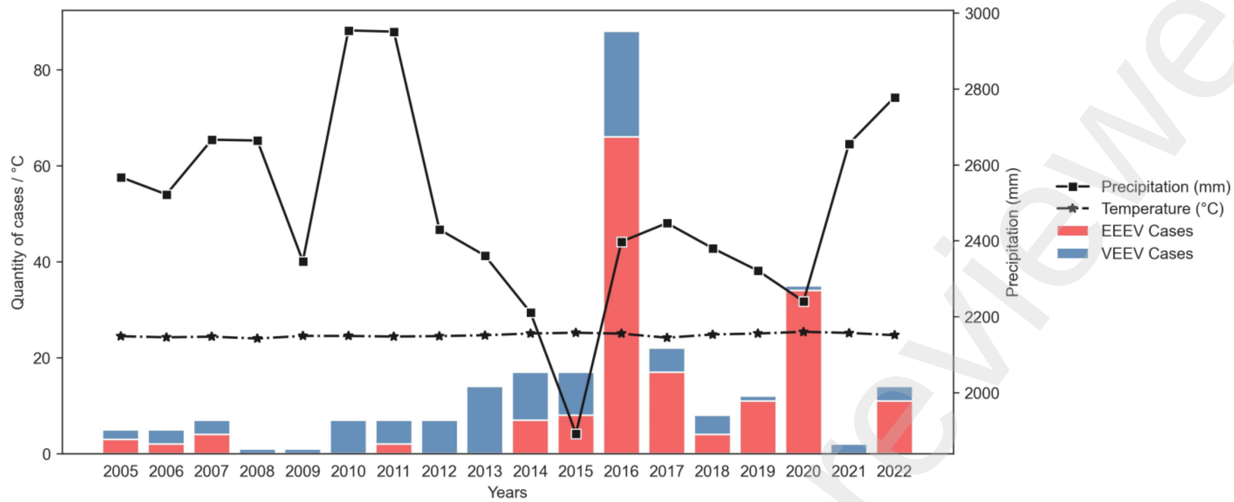

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Figure 2. Geographical distribution of equine infection by Eastern and Venezuelan equine encephalitis viruses-Colombia, 2005-July 2023.

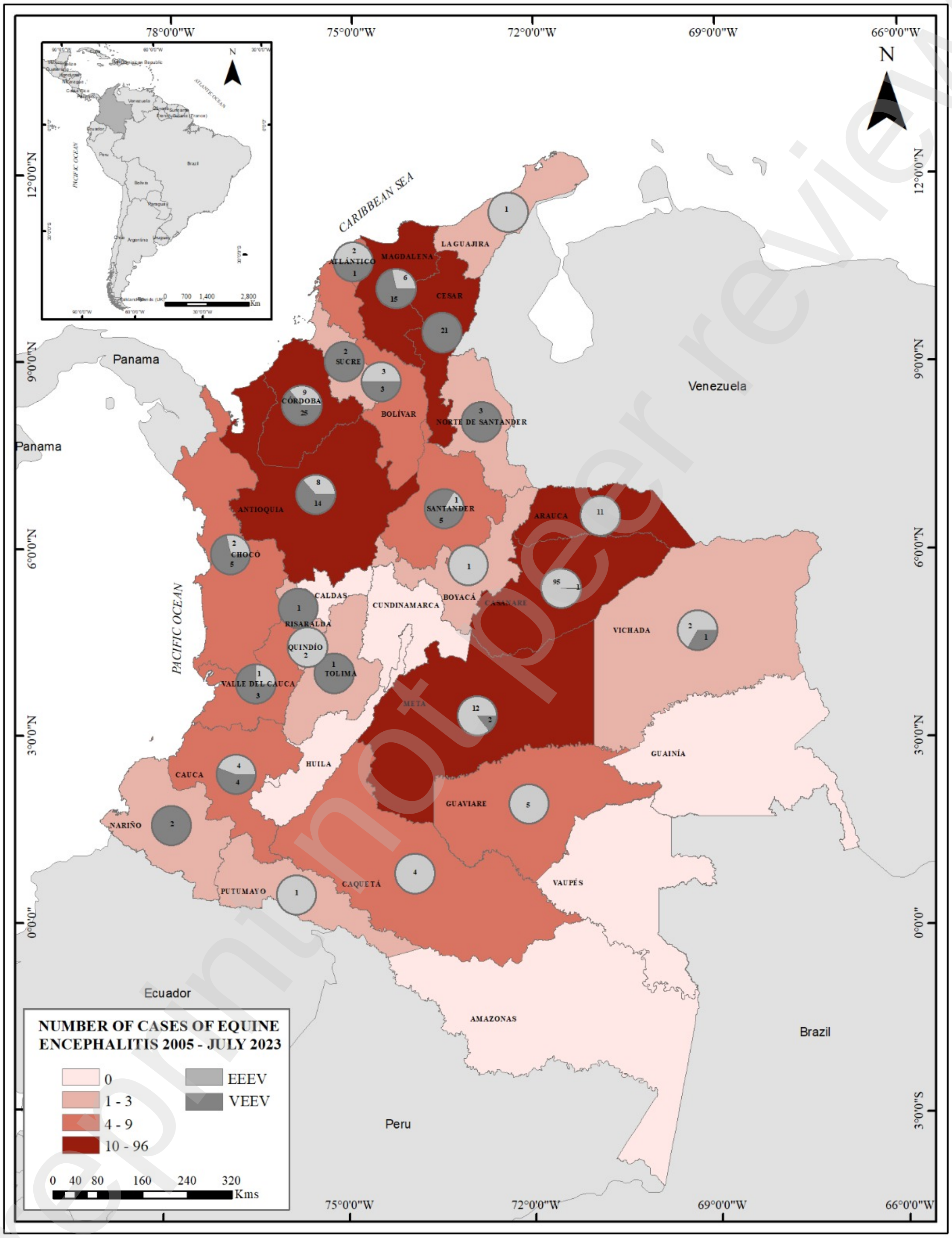

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Figure 3. Oceanic Niño Index and the dynamics of EEEV and VEEV cases, 2005-2023

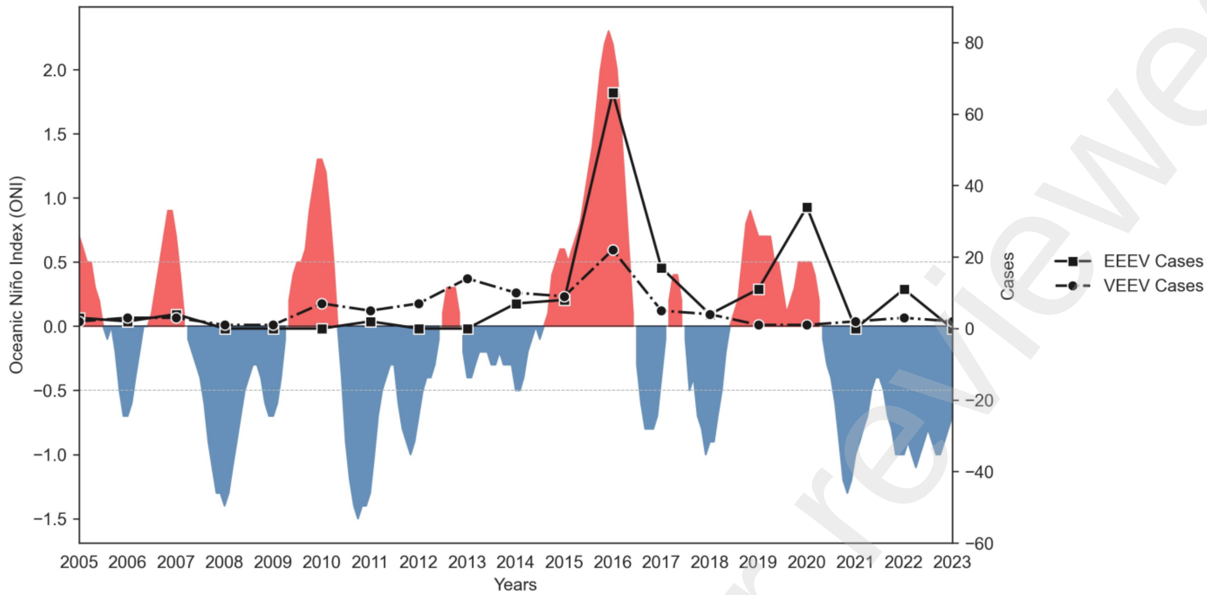

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Figure 4. Distribution of EEEV and VEEV in countries bordering Colombia and the rest of South America.

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Research in context

Evidence before this study. VEEV has been described in Colombia since 1962; agricultural institutions have monitored vaccination, which has allowed the control of viral transmission to equines and humans. However, there are no approved vaccines for the EEEV virus. In Venezuela, due to sociopolitical problems, a public health crisis has increased cases of vector-borne diseases. Cases of VEEV and EEEV lineage-Madariaga in humans have been reported. There are few molecular detections of encephalitis viruses in mosquitoes; in 2018, VEEV was detected in *Culex bidens*. Therefore, the circulation of this pathogen in mosquito species may represent a risk to human and animal health. Also, most clinical signs of equine encephalitis are confused with endemic arboviruses such as Dengue, Zika, and Chikungunya. **Added value of this study.** The solid epizootic activity of the EEEV and VEEV viruses between 2005-2023 shows that both viruses can become a human and animal public health problem. EEEV and VEEV are vector-borne diseases of public health concern that are likely to reemerge or spread due to climate change. Despite the absence of associations between EEEV and VEEV cases in equine and climatic variables, we know through research their influence on the dynamics of vector transmission of arboviruses. The research is pioneering for establishing possible prediction models at the departmental and national levels for future epidemics. **Implications of all the available evidence.** Based on the present work, countries such as Panama, Venezuela, Brazil, Peru, and Ecuador also must be alert to the increase in encephalitis cases in Colombia. Under the One Health concept, the National Institute of Health of Colombia could articulate with the Colombian Agricultural Institute and begin human and mosquito surveillance to prevent a possible outbreak.

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Table 1. Correlation coefficients of the association between temperature, rainfall, and the cases of EEEV and VEEV in Colombia.

Variables		**Pearson's correlation**	**P value**
Cases of EEEV	Temperature	0.41	0.09
	Rainfall	0.25	0.31
Cases of VEEV	Temperature	0.23	0.36
	Rainfall	0.23	0.35

*Correlation is significant at the 0.05 level

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